

WHISPERVC-AV: AUDIO-VISUAL CONTENT RESTORATION FOR NOISE-ROBUST WHISPER-TO-NORMAL VOICE CONVERSION

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ABSTRACT

Environmental noise obscures the content cues needed for whisper-to-normal conversion. We propose WhisperVC-AV, which restores acoustic content features while leaving the original WhisperVC conversion module unchanged. Its context-guided restoration module combines temporal acoustic context with synchronized lip features through attention and gated residual correction. Experiments on AISHELL6-Whisper show lower character error rates (CERs) than WhisperVC across three ASR systems, on clean speech and under all six signal-to-noise ratio (SNR) conditions with MUSAN noise. The largest gains occur at 0 dB SNR, where Qwen3-ASR CER falls from 36.29% to 28.88%. WhisperVC-AV also improves predicted speech quality while maintaining speaker similarity. The CER gains extend to unseen background noise without retraining, while visual controls support the use of utterance-specific lip cues. Audio examples are available on our demo page.¹

Index Terms— Whisper-to-normal voice conversion, audio-visual speech processing, content restoration, noise robustness

1. INTRODUCTION

Whisper-to-normal (W2N) conversion converts whispered speech into normal speech while preserving linguistic content and speaker identity. A speaker may whisper to avoid disturbing others, yet ventilation, household activity, or passing vehicles can remain audible. Whispered speech lacks periodic vocal-fold excitation [1], and noise further obscures its acoustic content cues. Therefore, W2N conversion must address both cross-phonation differences and noisy input.

Earlier W2N methods learn adversarial mappings [2], use variational autoencoders [3], or model source-filter synthesis [1]. More recently, WESPER and DistillW2N leverage self-supervised speech representations [4, 5], while WhispEar expands training data with pseudo-parallel whispered speech [6]. WhisperVC instead separates cross-phonation alignment from speech generation [7]. When noise corrupts its input features, the converter receives less reliable content information. This motivates us to restore the input representation while retaining the learned conversion mapping.

Synchronized lip motion is a useful source of complementary information because additive acoustic noise does not directly corrupt the video. This complementarity has been exploited in audio-visual speech enhancement [8], resynthesis [9], and representation learning [10, 11]. For speech recognition, V-CAFE uses visual-context attention and mask-plus-residual acoustic enhancement [12]. Similarly,

gated visual attention incorporates lip information into Whisper [13]. These approaches use visual evidence to strengthen acoustic representations for recognition.

Lip information has also been integrated directly into W2N conversion: W2N-AVSC trains a conditional VAE with a video-conditioned prior and sums acoustic and visual latent representations for decoding [14]. It evaluates audiovisual conversion with both clean and noisy whispered input. Rather than jointly learning audiovisual conversion, we use lip information to restore the acoustic representation supplied to a pretrained converter, preserving its learned conversion mapping. Our contributions are:

- We formulate **noise-robust W2N conversion** as **input-side content restoration**, reusing a pretrained converter without updating its conversion backbone.
- We develop a **context-guided audio-visual restoration module** that uses acoustic context to select nearby lip information and predicts a gated residual correction.

By restoring content at the converter’s input, we separate noise robustness from the learned whisper-to-normal mapping. This design enables us to reuse WhisperVC’s conversion capability while exploiting synchronized lip cues when acoustic evidence is degraded.

2. METHOD

2.1. Overview

Fig. 1 shows how we separate input restoration from conversion. For noisy whispered speech $\tilde{x}$, a frozen Whisper Large-V3 encoder fine-tuned on AISHELL6-Whisper [13] extracts acoustic content features H . Each utterance is instance-normalized over time per feature channel, with frozen affine parameters inherited from WhisperVC. Clean training targets are normalized independently. Our module combines H with synchronized lip features to produce $\hat{H}$ in the same acoustic space. WhisperVC’s alignment network maps $\hat{H}$ to normal-speech content features; its generator and vocoder reconstruct speech $\hat{y}$, conditioned on embedding $s(r)$ from same-speaker normal-speech enrollment r [7]. Only the restoration module is trained; the feature encoders and conversion backbone remain frozen.

2.2. Context-guided audio-visual restoration

In Fig. 2, a projection maps 1280-dimensional H to 256 dimensions. Three residual temporal convolutional layers, with kernel size 5 and dilations 1, 2, and 4, form acoustic context u . Frozen noise-robust AV-HuBERT [11], with audio disabled, extracts 1024-dimensional lip features from 25-fps video. Layer normalization and a learned

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¹ Demo page: larry-ziyue-yin.github.io/demo-whispervc-av/.

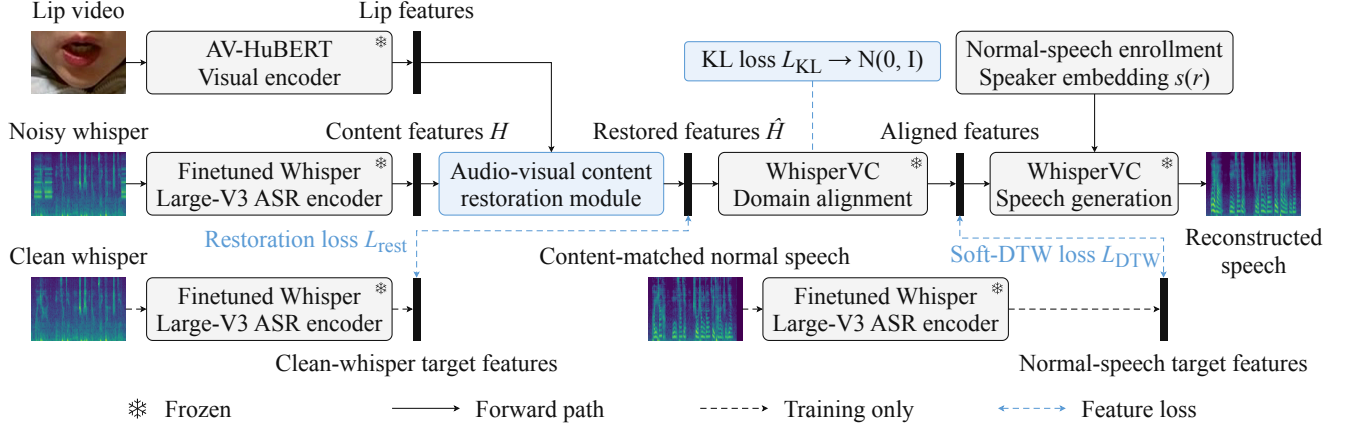


Fig. 1. Overview of WhisperVC-AV. Acoustic and visual cues restore content features while the original WhisperVC conversion module remains unchanged. All speech branches share the same frozen ASR encoder. Dashed paths and feature-loss connections are used only during training.

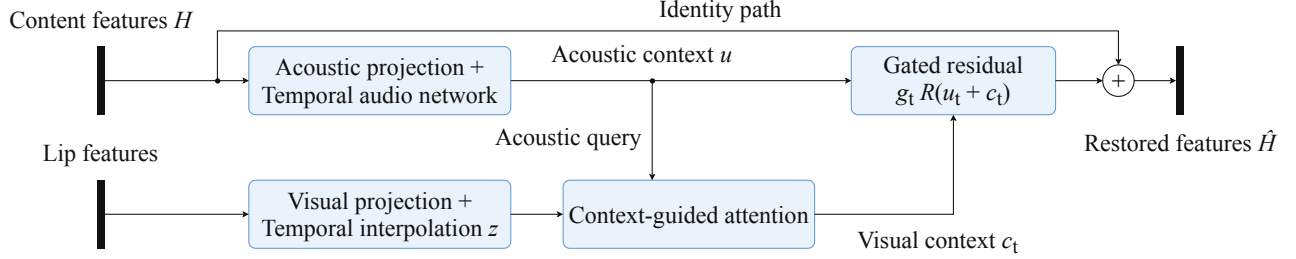


Fig. 2. Context-guided content restoration. Acoustic queries attend to projected lip features z within $\mathcal{N}(t)$. Acoustic context u_t and visual context c_t jointly determine the gated residual, which the identity path adds to content features H to obtain restored features $\hat{H}$.

projection map them to 256 dimensions; linear interpolation aligns them with the 50-Hz acoustic sequence, yielding z . Single-head scaled dot-product attention takes u as queries and z as both keys and values. Each query attends to $\mathcal{N}(t) = \{t - 4, \dots, t + 4\}$, spanning approximately ± 80 ms; invalid boundary positions are masked. Acoustic context u_t and visual context c_t jointly determine a scalar sigmoid gate g_t and the residual correction,

$$c_t = \text{Attn}(u_t, \{z_j : j \in \mathcal{N}(t)\}),$$

$$g_t = \sigma(W_g[u_t; c_t] + b_g), \quad \hat{H}_t = H_t + g_t R(u_t + c_t). \quad (1)$$

Here $[u_t; c_t]$ denotes concatenation, and R projects back to the input feature space. Zero-initializing R preserves the identity output.

2.3. Training and inference

Training pairs noisy whispered speech with its clean counterpart x . Noisy training inputs undergo waveform time masking with probability 0.5 before feature extraction; clean inputs are unmasked. We zero randomly placed spans of 0.2–0.6 s until 20% of waveform samples are masked, merging overlaps and truncating spans to the remaining budget. One masked variant is cached per noisy input. Feature-frame masks sample the waveform mask at 20-ms intervals to match the acoustic sequence. Restoration loss $\mathcal{L}_{\text{rest}}$ weights mean Smooth-L1 errors on masked and unmasked frames by 0.75 and 0.25, respectively, against clean features from the shared frozen encoder and normalization. Without masked frames, it uses the overall mean. Masks guide the loss but are not module inputs. For downstream compatibility, we retain WhisperVC’s alignment and latent objectives [7]. Soft-DTW [15] compares aligned features with content-matched normal-speech targets without assuming identical timing. KL regularization constrains the whisper-path posterior toward $\mathcal{N}(0, I)$.

Figure 1 shows the loss paths for the objective

$$\mathcal{L} = \mathcal{L}_{\text{DTW}} + \lambda_{\text{KL}} \mathcal{L}_{\text{KL}} + \lambda_{\text{rest}} \mathcal{L}_{\text{rest}}. \quad (2)$$

We set $\lambda_{\text{KL}} = 0.1$ to keep KL regularization modest and $\lambda_{\text{rest}} = 50$ to give the smaller, frame-averaged restoration loss an effective contribution alongside Soft-DTW. Although the alignment network is frozen, it remains differentiable: gradients from its Soft-DTW and KL losses pass through $\hat{H}$ to update the restoration module, while its own weights remain fixed. At inference, only noisy speech, lip video, and speaker enrollment are needed (Fig. 1).

3. EXPERIMENTAL RESULTS

3.1. Experimental setup

We retain AISHELL6-Whisper’s original partition boundaries [13] and use its video-covered subsets: 9,494 training utterances from 81 speakers, 2,235 validation utterances from 19 speakers, and the same 2,248 test utterances from 20 speakers, under clean input and each noisy condition. The utterance lists are available on our demo page.

All WhisperVC baseline metrics are recomputed on this subset; the original paper used a broader test set and Whisper-large-v3-turbo for CER [7]. FreeSound and SoundBible recordings from MUSAN’s noise subset [16] are mixed at 0, 5, 10, 15, 20, and 25 dB SNR, including environmental sounds such as crowd noise, dial tones, and squeaking doors. Music and speech subsets are excluded, and noise files are disjoint across partitions. Noise files and starting offsets are sampled separately for each SNR. Within each condition, all evaluated pipelines start from the same noisy recording of each test utterance. We also evaluate clean input.

Content-matched normal recordings from the same speaker provide training targets without frame-level alignment. The enrollment

Table 1. Character error rates (CER, %) on clean speech and under six SNR conditions with MUSAN noise. The cascades enhance waveforms before WhisperVC. WhisperVC-A is the separately trained audio-only ablation. Noisy avg. averages the six noisy conditions; lower is better, with each condition’s best result in bold.

ASR	System	Clean	0 dB	5 dB	10 dB	15 dB	20 dB	25 dB	Noisy avg.
Qwen3-ASR	WhisperVC [7]	11.47	36.29	26.23	18.91	15.40	13.43	12.34	20.43
	MP-SENet-DNS + WhisperVC	24.42	43.13	34.31	26.97	22.57	19.72	17.90	27.43
	SEMamba + WhisperVC	25.34	61.63	52.04	43.18	35.67	31.63	28.71	42.14
	Direct AV adaptation	10.40	29.99	21.51	16.18	13.20	11.65	11.05	17.26
	WhisperVC-A	10.11	30.92	21.76	16.08	13.27	11.66	10.87	17.43
	WhisperVC-AV (ours)	9.86	28.88	20.60	15.31	12.77	11.36	10.57	16.58
Finetuned Whisper (Large-V3) in [13]	WhisperVC [7]	13.68	52.97	36.36	23.92	17.70	15.99	14.55	26.92
	MP-SENet-DNS + WhisperVC	33.18	65.47	47.08	34.03	27.76	24.44	22.76	36.92
	SEMamba + WhisperVC	30.11	92.71	80.30	57.61	44.15	36.79	35.86	57.90
	Direct AV adaptation	12.24	41.36	26.28	18.31	15.18	14.17	12.85	21.36
	WhisperVC-A	11.46	33.46	23.15	17.36	14.52	12.94	12.28	18.95
	WhisperVC-AV (ours)	11.27	30.43	21.45	16.83	13.97	12.52	11.90	17.85
Paraformer	WhisperVC [7]	14.01	39.98	29.81	22.27	18.63	16.44	15.20	23.72
	MP-SENet-DNS + WhisperVC	28.13	46.68	38.16	31.04	26.25	23.22	21.27	31.10
	SEMamba + WhisperVC	29.57	64.10	55.21	47.25	39.71	35.72	32.81	45.80
	Direct AV adaptation	12.96	34.00	25.56	19.50	16.30	14.70	13.70	20.63
	WhisperVC-A	12.68	35.14	25.60	19.61	16.29	14.41	13.57	20.77
	WhisperVC-AV (ours)	12.33	33.20	24.73	19.10	15.80	14.10	13.23	20.03

Table 2. Mean CER reductions of WhisperVC-AV over six SNR conditions with MUSAN noise (percentage points), with pointwise 95% confidence intervals (CIs). All nine two-sided paired tests yield raw $p = 0.000005$ (199,999 label swaps; plus-one correction) and Holm-adjusted $p = 0.000045$.

ASR	Compared with	Reduction [95% CI]
Qwen3-ASR	WhisperVC	3.85 [3.57, 4.13]
	Direct AV adaptation	0.68 [0.48, 0.88]
	WhisperVC-A	0.85 [0.73, 0.97]
Finetuned Whisper (Large-V3) in [13]	WhisperVC	9.07 [7.76, 10.46]
	Direct AV adaptation	3.51 [2.64, 4.47]
	WhisperVC-A	1.10 [0.69, 1.54]
Paraformer	WhisperVC	3.69 [3.44, 3.95]
	Direct AV adaptation	0.60 [0.40, 0.80]
	WhisperVC-A	0.74 [0.64, 0.85]

speech, however, uses a different sentence spoken normally by the same speaker, during both training and inference. For evaluation, each test utterance uses the same enrollment recording across systems and SNR conditions. Targets and enrollment remain clean.

3.2. Evaluation metrics

Character error rate (CER) measures intelligibility and content preservation. Three ASR systems are used to assess the consistency of CER findings: Qwen3-ASR [17], Finetuned Whisper (Large-V3) in [13], and Paraformer [18]. For each condition, we sum character substitutions, deletions, and insertions over all test utterances and divide by the total number of reference characters.

Predicted output quality is assessed with DNSMOS P.835 [19], UTMOS [20], WVMOS [21], and NISQA [22]. Resemblyzer SECS, WeSpeaker [23], and WavLM [24] measure similarity to content-matched normal recordings; SpeechBERTScore [25] evaluates consistency with the same references.

3.3. W2N results and ablation analysis

We evaluate WhisperVC-AV against original WhisperVC [7] and examine waveform enhancement, direct visual adaptation, and the

contribution of the visual branch (Table 1). For the last comparison, we separately train an audio-only ablation, denoted WhisperVC-A. It removes the visual projection and attention paths in Fig. 2, so that the gated residual depends solely on acoustic context u_t . The acoustic network, supervision, and training schedule are retained.

The enhancement cascades apply magnitude–phase MP-SENet-DNS [26] or Mamba-based SEMamba [27] before WhisperVC, using official pretrained weights without task-specific fine-tuning. Both yield higher CER than WhisperVC on clean speech and under all six SNR conditions. MP-SENet-DNS, the stronger cascade, increases noisy-average Qwen3-ASR CER from 20.43% to 27.43%, whereas WhisperVC-AV reduces it to 16.58%. For these off-the-shelf enhancement cascades, waveform preprocessing does not improve W2N content preservation; task-specific feature restoration produces lower CER across the tested conditions.

W2N-AVSC’s trained model and audiovisual test data were unavailable to us for matched evaluation. Motivated by its audio-visual fusion strategy [14], we implement Direct AV adaptation within WhisperVC. This baseline trains gated visual fusion and the whispered-speech alignment branch with Soft-DTW and KL, using the same training subset and visual features; other modules remain frozen. It improves on WhisperVC, but WhisperVC-AV achieves lower CER on clean speech and at every tested SNR under all three scorers. This system-level comparison supports the effectiveness of restoring acoustic content while retaining the pretrained conversion module.

The WhisperVC-A ablation tests whether visual information improves on audio-only restoration. Adding the visual branch lowers noisy-average Qwen3-ASR CER from 17.43% to 16.58% and clean CER from 10.11% to 9.86%. Its largest gain occurs at 0 dB SNR, from 30.92% to 28.88%, compared with 10.87% to 10.57% at 25 dB SNR. Finetuned Whisper and Paraformer show the same A-to-AV improvement in every condition. Thus, synchronized lip cues contribute beyond the acoustic context, especially when noise makes the speech signal least reliable.

We assess six-SNR average CER gains using paired label-swap tests and 5,000 paired bootstrap draws, grouping each utterance’s six conditions. As shown in Table 2, gains over WhisperVC, Direct AV, and WhisperVC-A are significant under all three scorers after Holm

Table 3. Predicted speech quality, speaker similarity, and reference consistency for clean input and the average over six SNR conditions with MUSAN noise. Higher values are better; bold indicates the best rounded score within each condition.

System	Set	DNSMOS $\uparrow$				MOS predictors $\uparrow$			Speaker/reference similarity $\uparrow$			
		OVRL	SIG	BAK	P808	UTMOS	WVMOS	NISQA	SECS	WeSpk	WavLM	SBERT
WhisperVC [7]	Clean	3.07	3.51	3.72	3.58	2.83	3.29	3.13	0.83	0.68	0.94	0.78
	Noisy avg.	3.05	3.48	3.71	3.54	2.76	3.22	3.11	0.82	0.67	0.94	0.77
Direct AV adaptation	Clean	3.09	3.52	3.73	3.59	2.85	3.30	3.15	0.84	0.69	0.95	0.79
	Noisy avg.	3.08	3.52	3.73	3.58	2.82	3.28	3.14	0.83	0.68	0.94	0.78
WhisperVC-A	Clean	3.17	3.55	3.84	3.67	2.90	3.28	3.18	0.84	0.69	0.95	0.79
	Noisy avg.	3.16	3.55	3.84	3.67	2.89	3.28	3.18	0.83	0.69	0.94	0.78
WhisperVC-AV (ours)	Clean	3.16	3.55	3.83	3.67	2.90	3.28	3.18	0.84	0.69	0.95	0.79
	Noisy avg.	3.16	3.55	3.84	3.67	2.89	3.27	3.18	0.83	0.69	0.94	0.78

Table 4. W2N CER (%) with unseen background noise from DE-MAND at three SNR levels. Lower is better.

System	Qwen3-ASR			Finetuned Whisper (Large-V3) in [13]		
	0 dB	5 dB	10 dB	0 dB	5 dB	10 dB
WhisperVC [7]	24.72	17.26	13.78	30.80	23.61	18.24
WhisperVC-A	20.80	14.75	12.06	21.78	15.88	13.22
WhisperVC-AV (ours)	19.71	14.22	11.90	20.43	15.39	13.17

correction. The positive 95% confidence intervals (CIs) support consistent gains; the Qwen3-ASR visual gain is 0.85 percentage points, with a CI of [0.73, 0.97]. WhisperVC-AV improves on WhisperVC-A for 20, 19, and 20 of the 20 test speakers under the respective scorers. Resampling speakers gives 95% CIs of [0.59, 1.12], [0.63, 1.64], and [0.53, 0.98] percentage points. All remain positive, indicating that the visual benefit is not driven by only a few speakers.

Table 3 shows that the CER gains are accompanied by improved predicted speech quality. Relative to WhisperVC, WhisperVC-AV raises noisy-average DNSMOS OVRL from 3.05 to 3.16 and UTMOS from 2.76 to 2.89. DNSMOS, UTMOS, and NISQA also improve on clean input, while speaker/reference similarity scores are maintained or slightly higher. WhisperVC-A and WhisperVC-AV achieve comparable quality and similarity scores, while WhisperVC-AV further reduces CER. Thus, visual cues improve content preservation while retaining the quality gains of audio-only restoration.

3.4. Zero-shot noise generalization

Without retraining or adaptation, we evaluate the W2N performance of WhisperVC, the WhisperVC-A ablation, and WhisperVC-AV on whispered speech mixed with unseen background noise from DE-MAND [28]. We use kitchen (DKITCHEN), subway-station (PSTATION), and bus (TBUS) recordings at 0, 5, and 10 dB SNR, following the same noise-mixing protocol as MUSAN. WhisperVC-AV outperforms WhisperVC at all three SNRs under both ASR scorers (Table 4). Qwen3-ASR CER falls from 24.72% to 19.71% at 0 dB SNR and from 13.78% to 11.90% at 10 dB SNR. WhisperVC-AV also improves on the audio-only ablation at every SNR under both scorers, reducing three-SNR mean CER by 0.59 and 0.63 percentage points for Qwen3-ASR and Finetuned Whisper, respectively. These visual gains extend to unseen noise and are largest at 0 dB SNR, where acoustic evidence is most degraded.

3.5. Visual-input controls

We probe WhisperVC-AV’s use of visual information with parameters and acoustic input fixed. Same-speaker mismatch preserves identity but replaces the utterance’s lip movements, while shifts of ± 10 native

Table 5. Qwen3-ASR CER (%) for visual controls on clean speech and under six SNR conditions with MUSAN noise. Lower is better.

Video	WhisperVC-AV (ours)						
	Clean	0 dB	5 dB	10 dB	15 dB	20 dB	25 dB
Correct	9.86	28.88	20.60	15.31	12.77	11.36	10.57
Disabled	10.22	31.12	22.14	16.27	13.37	11.54	10.69
Mismatch	10.15	31.78	22.63	16.70	13.57	11.78	10.81
Shift -10	10.29	31.34	22.25	16.50	13.59	11.52	10.92
Shift $+10$	10.25	31.59	22.06	16.33	13.44	11.70	10.89

frames (approximately ± 400 ms), with boundary padding, alter temporal correspondence. Unlike the separately trained WhisperVC-A ablation, the disabled-context control retains WhisperVC-AV’s trained parameters and sets $c_t = 0$ only at inference.

Every control increases Qwen3-ASR CER for clean speech and all six SNR conditions (Table 5). At 0 dB SNR, CER rises from 28.88% to 31.12% without visual context and to 31.78% with same-speaker mismatch. Mismatch is also worse in every other noisy condition: incorrect lip content can be more harmful than no visual context. Both temporal shifts also increase CER across conditions. Together, these controls show that useful visual input must carry the utterance’s articulatory content: retaining the correct speaker alone is insufficient. The separately trained audio-only ablation establishes the benefit of adding vision, while these fixed-model controls establish the importance of the supplied lip content. Their agreement supports the use of synchronized visual evidence to complement acoustic restoration, especially when noise obscures speech cues.

4. CONCLUSION

In this paper, we propose WhisperVC-AV to restore acoustic content while leaving the original W2N conversion module unchanged. It achieves the lowest CER among evaluated systems, with consistent gains across three ASR scorers on clean speech and under all six SNR conditions with MUSAN noise. The model also improves predicted speech quality over WhisperVC while maintaining speaker similarity. Visual cues significantly reduce noisy-average CER over the audio-only ablation. The full system also generalizes to unseen noise without retraining. These findings motivate low-volume and assistive communication applications using synchronized lip video and clean enrollment. Future work includes uncontrolled video, reverberation, noisy enrollment, other languages, and human listening tests.

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6. COMPLIANCE WITH ETHICAL STANDARDS

This study uses existing AISHELL6-Whisper recordings in accordance with the data provider's authorization. No new participants were recruited or human-subject data collected for this study. No additional ethical approval was required for this secondary analysis.

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